

Lenses That Summarise: Bearing-Faithful Aggregating Map Lenses, from a Single Lens to a Gridded Glyphmap [TODO working title]

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Working draft — September 25, 2026

Abstract

A map lens that *summarises* the data under it—rather than magnifying or filtering it—must decide what to summarise and where to put the summary. Existing lens models leave both inside a single lens function and join. We refine them into seven stages (selection, binning, normalisation, placement, anchor, marks, association) and show that a category ring, a bearing-faithful necklace, a route profile and a gridded glyphmap are points in one space, reached by changing arguments. The part of the space we find new is where the lens boundary itself carries data: aggregates placed on a ring at the bearing of what they summarise, displaced minimally rather than dropped; a leader that appears exactly where placement or unrolling has given adjacency away; a ring that unrolls into an axis without re-solving; within-unit structure read from the lens’s own polar frame; and a radius control that shows how fragile the reading is. We relate the model to lens design spaces, focus-region labelling, necklace maps and geographically weighted statistics, state which parts we concede to them, describe an open-source implementation, and set out the controlled study that the central design claim still requires. [TODO Rewrite once the study has run; keep to the venue’s word limit.]

1 Introduction

Interactive lenses are one of visualization’s most general interaction primitives: a movable region under which the base visualization is shown differently [6, 58, 59]. On maps, many lenses *magnify*, *filter* or *re-render* what lies beneath them. Others instead *summarise* it: a probe pairs a region with charts of its contents [13], a time lens wraps a query circle in a ring of temporal aggregates [57], and a shipped map-lens product draws a bar chart of the data under each lens around its rim [62].

A summarising lens has to answer two questions that a magnifier does not. What is summarised—which members, decomposed how, normalised against what? And where does the summary go—in which order around the

lens, on what curve, and how is each mark tied back to the ground it describes? In the rings we have seen, the second question is answered by habit: marks are laid out in category (or time) order, so the most valuable channel on a ring—angular position—carries a category or time order and nothing geographic. A bar at eleven o’clock says nothing about the north-west.

This paper makes three moves. First, we refine Tomin-ski et al.’s lens model (selection σ , lens function λ , join $\bowtie$ [59]) for summarising lenses into seven stages, $\lambda = \mu \circ \nu \circ \beta$ and $\bowtie = \rho \circ \gamma \circ \varphi$ (section 3), and use it to describe existing techniques (section 4) and to generate new ones (section 5). Second, we make angle mean *bearing*: each aggregate is placed on the lens boundary at the bearing of what it summarises, using necklace-map placement [53] with minimal displacement; a leader is drawn exactly where the gap between a mark and its data would mislead; and because placement is solved in the curve’s parameter, the ring unrolls into an axis without re-solving. Third, we treat the number of lenses as a parameter, so that one lens, a handful of small multiples and a gridded glyphmap are the same object, and we are explicit about how much of that continuum is geographically weighted statistics under another name (section 8).

Contributions.

- A seven-stage model of summarising lenses that refines the selection–function–join model, with two values the existing lens design spaces lack: an effect scope *on the boundary*, and thematic marks under a spatial layout (section 3).
- Bearing-faithful placement of aggregates on a lens boundary, with a residual leader and curvature as a free axis (sections 3.5 and 3.8).
- Within-unit structure read from the lens’s polar frame, and the elasticity of the count with respect to the radius drawn on the radius control, positioned against the local point-pattern statistics it re-expresses (sections 3.9 and 3.10).

- The lens-to-glyphmap continuum as one object (section 5.1).
- An open-source implementation (section 6) and the design of the controlled study the central claim requires (section 7).

[**TODO** Teaser figure: category ring $\rightarrow$ necklace $\rightarrow$ unrolled with leaders $\rightarrow$ field, from **figures/**.]

2 Related work

The closest precedents to a bearing-faithful summarising lens are mostly not called lenses. They sit in five literatures: interactive lenses and their design spaces; labelling around a focus region; off-screen indicators on the viewport border; radial and necklace cartography; and glyph maps with their underlying local statistics.

2.1 Interactive lenses and their design spaces

Magic lenses were introduced as movable see-through filters that modify the presentation of the objects beneath them [6]; a taxonomy of see-through tools followed [7], and composing overlapping lenses was addressed as a programming-model problem [24]. Tominski et al.’s surveys model a visualization lens as a pipeline attached to the visualization pipeline through a selection σ and a join $\bowtie$, with a lens function λ between them (the symbol is introduced in the extended version), classify lenses by data type and task, and add an *effect extent*—lens interior, side effects, separate view—to earlier axes [58, 59]. Among their future directions are flexibly combining lenses and a unified lens toolkit [58, 59]; both are directions this work speaks to. Mota et al. derive a seven-dimension design space for spatially embedded lenses from 45 papers (2006–2020): position, orientation and scale; shape; dimensionality; effect scope (interior, exterior, separate view); effect imagery; effect encoding (thematic or spatially based); and viewpoint dependency [45]. Two of these dimensions have no value for a necklace ring: its effect sits neither inside, outside nor apart from the lens but *on its boundary*, and its encoding is thematic marks under a spatial layout.

Lenses on maps have mostly magnified, filtered or re-rendered: JellyLens adapts its shape to the features under it [48]; MoleView pushes elements outside an attribute range towards the lens border [34]; ClusterLens re-aggregates points inside the lens at a finer resolution [70]; route detail lenses place magnified views around the map border in route order, with leader lines only to the first and last [37]; and the Sampling Lens samples points inside a region to reduce clutter, listing aggregate statistics as future work [20]. VectorLens selects curves by position, diameter, direction and angular tolerance, and draws the selected angular ranges on a

ring around the lens—a ring whose angle means direction, used for selection rather than display [18]. The name GlyphLens already belongs to a view-dependent lens that pulls occluding 3D glyphs aside [60].

Closer to our subject are lenses and probes that *summarise*. Probes pair a user-defined region with a pane of local charts, linked to it by a line or a shared colour and placed by the user [13]; a follow-up alerts analysts when regions being compared differ unfairly on some dimension, and offers semi-automatic boundary edits, while explicitly setting the scale component of the modifiable areal unit problem aside [14]. TrajectoryLenses combine map lenses with set operations and show aggregated attributes in the vicinity of the lens or in colour-matched panels [39]; a traffic-flow selection widget filters by area and direction and shows the selection in annotation windows [50]; GTMapLens browses geo-tagged text with movable lenses [42]. The closest academic precedent we found is the *time lens* of Tominski et al.: a circular query area on a trajectory map, with a ring segmented into cyclic time bins whose fill shows the trajectories inside the circle [57]. There, angle means time. Outside the literature, VisQuill ships map lenses that aggregate the data beneath them into a bar chart around the rim [62]; its documentation does not say what angle encodes, and its open-source lens blueprint spaces one bar per category evenly along the arc. [**? Several of these readings are from abstracts only (TrajectoryLenses, GTMapLens, the traffic-flow widget); the time lens’s placement around the query circle is strongly implied but not stated in the text we read; see the audit.**]

2.2 Marks around a focus region

Excentric labelling shows the labels of objects inside a cursor-centred circle, connected to them by lines and updated as the cursor moves [21]. Labels are laid out in left and right stacks; in its radial variant bearing is used only to order them. When there are too many objects it shows a count and a sample of labels, and a glyph “summarizing the contents of the area” is proposed as future work [21]. *Extended excentric labelling* adds support for high and uneven density and for summary statistics [5] [**? what those statistics are and how they are drawn is unverified: full text not accessed**].

The algorithmic literature on *focus-region* and *contour* labelling places labels on the boundary of a circular focus itself [4, 22]. Fink et al. connect each label to its site by a straight or Bézier leader; in their radial model a label’s port is the radial projection of its site onto the circle, and sites too close in angle are resolved by keeping a maximum conflict-free subset; a facility-location variant clusters the sites and labels one representative per cluster; and the focus region’s position can itself be optimised [22]. Haunert and Hermes require every leader to point at the centre, allow one position per

label, and solve the resulting maximum-weight independent set in real time in a browser [32]. Boundary labelling has been extended to moving focus regions [33], to dense point sets explored at a fixed map scale with labels placed along the map’s lower edge [26], to fisheye displacement of labels out of a focus with optimisation once it stops [46], and to labels drawn as arcs in an annular orbit around a circle with minimum total leader length [9], where a study with 54 participants found straight leaders faster than orbital-radial ones at similar accuracy [63]. [\[? Wallinger et al.’s numbers were read from arXiv v1, not the PacificVis version of record.\]](#)

This is the geometric twin of what we do, and we build on it rather than around it. The difference is in what the marks are and how conflicts are resolved. In every one of these techniques a label stands for one object, a leader is always drawn, and a conflict with the bearing is resolved either by *selection* (dropping a label) or by freeing the label’s position. A summarising lens places *aggregates*, keeps every one of them, displaces them *minimally* from their bearing, and draws a leader only where the resulting gap is large enough to mislead—so the leader’s length is the residual. We did not find a labelling technique that draws leaders conditionally on displacement.

Off-screen indicators. The same geometry appears at the edge of the viewport. Off-screen techniques such as Halo, City Lights, EdgeRadar, Wedge and Ambient Grids draw proxies at or near the viewport border for objects beyond it, and all of them encode those objects’ direction; EdgeRadar and Ambient Grids compress the off-screen space into a dedicated border region, and Ambient Grids aggregates it into grid cells [2, 30, 31, 35, 36, 69]. How a proxy’s position should relate to its object is not settled. In a study with 18 participants on a rectangular border, 94.4% of localisation judgements assumed an orthographic rather than a radial projection, with lower error for orthographic [36]; the authors note that radial projection is the familiar one when there is a point of interest, as in car navigation. A lens boundary is circular and its centre is exactly such a point, so the result transfers only in part; we return to it in section 7. [\[? The off-screen techniques other than Jäckle et al. 2017 are cited from their registry records and from Jäckle et al.’s classification, not from their own texts.\]](#)

CompaRing copies a handful of objects into the slots of a ring for comparison and gives each slot an arc that points towards the object’s original location, wider for objects further away [56]; laying the slots out by the objects’ directions was suggested by the experts it was shown to, but not built. Ring-shaped maps have also been proposed for egocentric navigation in augmented reality [16].

2.3 Radial and necklace cartography

Radial layouts are surveyed by Draper et al. [17]. The direct ancestor of bearing-faithful placement is the *necklace map*: the regions of a map are projected onto intervals of a curve that surrounds them, and proportional symbols are placed without overlap inside their intervals [53]. The curve may be any star-shaped curve; a region’s interval can be a fixed-width window around the bearing of its centroid or the wedge it subtends from the centre; and several nested or disjoint necklaces may share a map. The objective is to maximise a common symbol scale, after which symbols are pushed towards the middle of their intervals by a force-based step. For a fixed order this is solvable in $O(n \log n)$; with arbitrary order it is NP-hard for wedge intervals, and ordering symbols by the bearing of their regions is *not* always optimal for that objective [54]. Necklace maps explicitly do without leaders, relying on proximity and colour, and acknowledge that the association between a symbol and its region is weaker than on other maps—“Interactivity can help to strengthen this association” [53].

Ring maps arrange evenly spaced “spokes”, one per areal unit, in rings of attributes around a base map, and tie every spoke to its unit with a leader line; their authors name the limited representation of spatial topology in the rings as the main limitation and point to interactive ring maps as the remedy [55].

A necklace lens differs from both in three ways. It is interactive and local—a movable selection rather than a whole map. Its placement minimises squared displacement from each mark’s preferred bearing at a mark size fixed by the ring, rather than maximising a common scale within intervals. And its leaders are conditional: drawn where the gap between mark and data is large, absent where adjacency suffices—a middle course between the necklace map’s no leaders and the ring map’s leader for every spoke.

2.4 Glyph maps, within-cell structure and local statistics

Glyph placement strategies divide into data-driven and structure-driven [64]; necklace placement is structure-driven layout with a data-driven preferred position. Glyph design guidance is well surveyed [10], but controlled evidence on glyph layout and on glyphs over maps is thin [25]. Glyph-maps place small temporal glyphs at the data’s own grid locations [65]; *tilemaps* and *gridded glyphmaps* aggregate into a regular grid and draw a multivariate glyph per cell [40, 51, 52], and multivariate glyph placement with level of detail uses hierarchical centroids [44]. Tilemaps were proposed as a way to summarise the outputs of geographically weighted statistics, with the grid interactively resized and panned to expose the modifiable areal unit problem and a distance-decay kernel larger than the tiles as an

option [51]. Gridded glyphmaps keep the grid fixed in screen space and re-aggregate on zoom, and name glyphs that convey within-cell heterogeneity as future work [52].

That heterogeneity is what two hexagon techniques encode. Honeycomb Plots observe that colour-coded hexbins cannot distinguish “uniform distributions within tiles . . . from clusters or trends if their numbers of points match”, and cut each tile along the regression plane of its within-tile density; a between-subjects study with 42 participants found strong evidence for slope tasks [61]. HexTiles derive a per-tile confidence from weighted within-tile variance, argued as mitigation of the modifiable areal unit problem [38]; its weighting is described only in words, and its user study (seven participants) tested the tiles without the variance encoding. Polar histograms of street bearings summarise a city’s orientation order [8]; they bin the bearing *of* edges rather than bearing *from* a chosen centre, which is the quantity a lens bins.

Local statistics and scale. A disc-shaped lens evaluated at many centres is a moving-window statistic. Geographically weighted summary statistics compute local summaries with a spatial kernel around each location [11], in the family that began with geographically weighted regression [12]. Geographically weighted visualization makes them interactive, calls the GW mean “simply a moving spatial window mean smoother”, plots summary statistics across bandwidths as a *scalogram*, notes that the statistics “could . . . be centred on any points, such as those comprising a regular grid”, and adds *directed* weighting, in which a kernel is sharpened towards a principal direction at thirty-degree clock points [19]. Local correlation has been shown varying with scale and geography [29], and local indicators of spatial association decompose global statistics into per-observation contributions [1]. The modifiable areal unit problem—results that change with the scale and the zoning of units—is reviewed by Openshaw [47] and shown to be unpredictable in multivariate analysis by Fotheringham and Wong [23]. For point patterns, Ripley’s K counts events within growing distances of an arbitrary event, with $K(t) = \pi t^2$ under a Poisson process and simulation envelopes for inference [49]; second-order neighbourhood analysis takes the same view from each individual point [27]; and ring (O-ring) statistics complement the cumulative circle [66]. The log-log slope of mass within distance of a chosen centre is also the mass-radius dimension of fractal urban analysis [68]. [? Getis & Franklin and Wiegand & Moloney were read at abstract level only; a more canonical citation for radial (mass-radius) analysis should replace or join Yamu et al., whose main subject is planning theory. Brunsdon et al. 2002 was read at abstract level; whether it discusses a uniform kernel is unverified.]

These are the statistics a summarising lens computes, and we claim none of them. Directed GW weighting in particular is a statistical precedent for decomposing a neighbourhood by direction. What the lens contributes is a *placement* of that decomposition: each direction’s summary is drawn at its bearing on the lens boundary, and the radius control shows how the reading depends on the radius before the analyst drags it.

3 A model of summarising lenses

3.1 From selection, function and join to seven stages

Tominski et al. model a lens as a small pipeline embedded in the visualization pipeline: a selection σ picks out the part of the data the lens affects, a lens function λ transforms it, and a join $\bowtie$ composes the result with the base visualization [58, 59]. The model is deliberately general, and for the lenses it was built around—magnifiers, filters, restylers— λ is a transformation of the selected elements and $\bowtie$ is compositing into the lens’s interior. Neither needs further structure.

A *summarising* lens is different in both places. Its λ does not transform the selected elements; it replaces them with a much smaller derived dataset—one value per bin—and then with marks for those values. And its $\bowtie$ is no longer compositing: the marks have to go *somewhere*, and where they go carries meaning. We therefore refine the two opaque stages into three each, and keep σ :

$$\lambda = \mu \circ \nu \circ \beta, \quad \bowtie = \rho \circ \gamma \circ \varphi. \quad (1)$$

Binning β decomposes the selected members; normalisation ν turns each bin into a value against a baseline; marks μ turn values into glyphs. Placement φ decides where on a curve each glyph goes; the anchor γ decides what that curve is; association ρ decides how a glyph is tied back to the ground it summarises. Figure 1 shows the seven stages and the correspondence. The rest of this section defines each stage precisely enough that the techniques of section 4 can be written as paths through it.

3.2 Selection σ

A selection is a region R with an anchor point c from which bearing and distance are measured. It returns the members $M = \{m_i\}$ inside R , each annotated with its distance d_i and bearing θ_i from c —or, for a corridor along a polyline P , with its chainage s_i (distance along P) and signed offset o_i (distance across, positive to the left of travel). We distinguish five shapes: a *disc*, an *annulus*, a *sector*, a *corridor*, and an arbitrary *polygon*. A lasso, an administrative unit and an isochrone are all polygons: they differ only in where the shape came from,

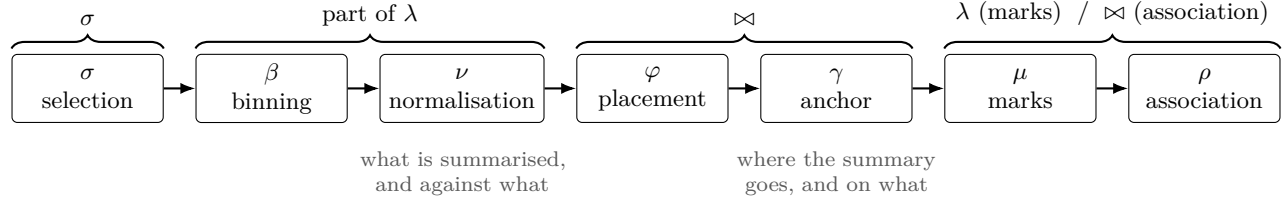


Figure 1: The seven stages of a summarising lens, and how they refine Tominski et al.’s selection σ , lens function λ and join $\bowtie$ [58]. The execution order interleaves the two refinements (marks are sized before they are drawn on the anchor), which is why $\lambda = \mu \circ \nu \circ \beta$ and $\bowtie = \rho \circ \gamma \circ \varphi$ are written as a factorisation rather than as two contiguous blocks.

so a lens that accepts rings accepts all three. A polygon has no natural centre, so c defaults to its area-weighted centroid and can be overridden (an isochrone’s origin is the obvious override).

For areal data—census units rather than points—a member can be partly inside R . Two standard answers are supported: count a unit if its centroid is inside, or weight it by the fraction of its area inside. The measure being aggregated must then declare whether it is *extensive* (counts, which apportion and sum) or *intensive* (rates, shares and medians, which do neither and are averaged with a weight). Nothing about a number says which it is, and summing a column of percentages produces a plausible and wrong picture, so the implementation requires the declaration.

3.3 Binning β

Binning partitions M into bins $B_1, \dots, B_K$ and gives each bin a *preferred position* $p_k \in [0, 1)$ on the anchor curve and, optionally, a *feasible interval* $I_k \subseteq [0, 1)$. These two quantities are the only things placement consumes. Six modes are defined:

categorical one bin per category; p_k is a nominal slot, so angle carries order and nothing else.

angular one bin per bearing sector; p_k is the sector’s bearing over 360° , so angle *is* bearing.

radial one bin per distance band; there is no bearing, so bins take nominal slots and the radial axis is the reading.

cross bearing sector $\times$ category.

chainage one bin per stretch of a corridor; $p_k = s/L$ on an open curve.

unit one bin per areal unit; I_k is the arc the unit subtends seen from c , which is Speckmann and Verbeek’s original necklace primitive [53].

A categorical bin may also be positioned at the circular mean of its members’ bearings, which is what morphs a category ring into a bearing-faithful one (section 5). That position is only meaningful when the members are

concentrated, so every bin also carries the mean resultant length $R_k \in [0, 1]$ of its bearings [43]; we return to this in section 3.9.

3.4 Normalisation ν

Normalisation maps a bin to a value v_k . *Count* and *density* (count per unit area) are the familiar pair; a radius slider silently confounds them, which is why normalisation is a stage and not a formatting option. *Share* is n_k/n . The *location quotient* $LQ_k = (n_k/n)/(b_k/b)$ and a z -score compare the lens against a baseline profile b ; *delta* is a signed difference from a reference (another lens, time or region)—in the terms of visual comparison, an explicit encoding of the difference rather than juxtaposition or superposition [28]. The baseline is itself a choice—the complement of the lens, the study area, or a fixed reference—and changes the reading materially. One lens defaults to its complement (“what is this place unusually full of, compared with its surroundings”); a field of lenses must share one baseline, or every cell becomes average by construction (section 5.1).

3.5 Placement φ

Let mark k need half-width h_k on the curve (its own width plus any room reserved for its label, in units of the curve parameter), and let $\delta(x, y)$ be the cyclic difference on $[0, 1)$. Placement solves

$$\begin{aligned} \min_x \quad & \sum_k w_k \delta(x_k, p_k)^2 \\ \text{s.t.} \quad & \text{the arcs } [x_k - h_k, x_k + h_k] \text{ are disjoint, } x_k \in I_k, \end{aligned} \tag{2}$$

the necklace-map placement problem of Speckmann and Verbeek [53, 54] with a squared-displacement objective. For a fixed cyclic order and no intervals, substituting $y_k = x_k - c_k$, where c_k accumulates the minimum spacings $h_{k-1} + h_k$, turns non-overlap into monotonicity of y , so the optimum is a weighted isotonic regression, solved exactly in $O(K)$ by pool-adjacent-violators. The implementation orders marks by preferred position, tries each of the K places to cut the cycle, and keeps the cheapest, for $O(K^2)$ overall. [?] The implementation assumes that

ordering marks by preferred position is optimal (“any crossing solution can be uncrossed without increasing cost”). That is standard for equal widths but unproven here for unequal h_k and w_k . For the related max-scale necklace problem with centroid intervals, Speckmann and Verbeek show that the order by region bearing is *not* generally optimal (it is a tight $\frac{1}{2}$ -approximation) [54]; their objective and constraints differ from eq. (2), so this is a warning, not a counterexample. A proof or a counterexample search is needed.]

Two parts of the solver are approximate, and we say so rather than hide it. When the marks nearly fill the cycle, the wrap-around constraint is enforced by alternating projection between monotone sequences and sequences of bounded span, which converges to a feasible point but not, in general, to the projection itself (Dijkstra’s algorithm would); and feasible intervals are enforced by clamping and re-solving for a fixed number of passes. Both are exact when inactive, which is the common case for a lens with a few dozen bins. **[TODO Either implement Dijkstra’s algorithm for the bounded-span case and an exact interval method, or keep the approximation and report its gap on a benchmark.]**

Two placements are degenerate cases of eq. (2): *block* placement, in which p_k are nominal slots with fixed spacing and the solver has nothing to do, and *stacked* placement, in which each variable gets its own concentric curve and competes only with itself, so displacement vanishes at the cost of radial space and cross-variable comparability.

3.6 Anchor γ

Placement is solved in the curve’s parameter, and each mark’s room is reserved as a *fraction* of the curve. Nothing in it assumes a circle. Any curve of the same length ℓ accepts the same solution, so the anchor is a free axis, and the useful thing to vary along it is curvature. For a ring of radius r_0 ($\ell = 2\pi r_0$) we define the family γ_u , $u \in [0, 1]$: the arc of length ℓ and curvature $(1 - u)/r_0$ through a fixed parameter t_0 , rotated so that γ_1 is horizontal. $u = 0$ is the closed ring; $u = 1$ is a straight baseline on which bearing becomes a horizontal axis. Changing u never re-runs placement: it is a change of anchor, not a re-solve, so it composes with every other stage. The seam falls opposite t_0 ; choosing t_0 in the widest gap between marks means opening the ring never cuts a mark in two.

The same axis applies to an open curve. Straightening a corridor onto its own chainage produces a route profile, which is a *linear cartogram*: chainage and offset survive exactly, position does not.

The trade the anchor controls is exact and worth stating. A ring keeps *adjacency*—each mark points at the part of the map it summarises—and pays for it in comparability, because every mark grows from its own

baseline in its own direction. A straight axis makes lengths directly comparable and gives labels somewhere to go, and pays in adjacency.

3.7 Marks μ

A mark turns v_k into a glyph at $\gamma(x_k)$: a radial *bar* (length $\propto v_k$), a *disc* (area $\propto v_k$, the classic necklace-map symbol), or a *rose* (the bin’s own angular histogram, oriented to true north so that roses compare with each other and with the compass). Which way a mark grows is a second, weaker way of buying comparability: along the curve’s outward normal (maximal association), always screen-up (one shared baseline; on a closed ring the lower marks grow back across the lens), or upright but never inward (a shared axis with two baselines). A nested glyph such as a rose carries two readings and must say which owns size; roses default to equal size, because a rose scaled by value is illegible for exactly the small bins whose direction is least certain.

3.8 Association ρ

On a closed ring around its own selection, adjacency associates a mark with its ground for free. Two things spend that: *placement*, which moves a mark off its bearing to avoid overlap, and the *anchor*, which under unrolling detaches the whole chart from the geography. They are the same loss by degrees, so one encoding answers both. A *leader* runs from the mark to the position placement tried to honour—the preferred bearing—at the members’ own mean distance from c , drawn in the lens’s azimuthal frame. Its length is therefore the association that was given away.

The leader is triggered by the *gap* between where a mark is and where its data is, not by the solver’s reported displacement. The distinction matters under block placement: the solver is asked for a nominal slot, reports zero displacement, and yet every mark is as far from its data as it can be. With the gap as trigger, morphing a ring from bearing back to category order fades the leaders in exactly as the marks leave their bearings. A leader is refused where it would have to guess: a distance band or a nominal slot has no bearing to point along, and a lens without a metric scale has no distance to point at.

3.9 Within-unit structure

Every stage above yields one number per bin, and that number hides whether the bin’s members are spread evenly or piled in one corner—which is what makes an aggregate fragile to how its unit was drawn. Hexagon-based techniques have begun to encode this: Honeycomb Plots cut each tile along the regression plane of its within-tile density, a “diamond cut” [61], and HexTiles derive a per-tile confidence from weighted within-tile variance,

argued explicitly as mitigation of the modifiable areal unit problem [38].

A lens gets this more cheaply than a tile, for a structural reason. A hexagon has no privileged origin or orientation, so the within-tile distribution must be fitted. A lens is already a polar coordinate system centred on a point the analyst chose, so its within-unit distribution decomposes, with no fitting, into bearing and distance. For each bin we compute the circular mean $\bar{\theta}_k$ and mean resultant length R_k of member bearings, the circular standard deviation $\sqrt{-2 \ln R_k}$ [43], and the mean, coefficient of variation and edge share of member distances; on a corridor, the signed mean offset (*bias*) and the mean offset sign (*sidedness*), which disagree precisely when provision is balanced by count but not by distance. These are drawn as a *spread* arc (circular SD), a *gradient* arrow (lens-level $\bar{\theta}$, length $\propto R$), the members drawn back through the aggregate (*inclusions*, after Honeycomb’s “amber inclusions”), or the rose mark.

3.10 The radius control as a scale diagnostic

The unit of a lens is modifiable by a control the analyst is already holding, so the lens can report how much its reading depends on that choice. Let $N(r)$ be the number of members within distance r of c . We define the *elasticity* of the count with respect to the radius,

$$E(r) = \frac{d \ln N}{d \ln r} = \frac{r N'(r)}{N(r)} = 2 \frac{\lambda_{\text{rim}}(r)}{\bar{\lambda}(r)}, \quad (3)$$

where $\lambda_{\text{rim}} = N'(r)/2\pi r$ is the intensity on the rim and $\bar{\lambda} = N/\pi r^2$ the mean intensity inside. The last form follows directly and is the useful reading: $E = 2$ when the rim is as dense as the interior (uniform density), $E \rightarrow 0$ when everything is already inside, and $E \gg 2$ when a cluster sits just beyond the rim and the reading is about to jump. The implementation uses a backward difference over a band of relative width b ,

$$\hat{E}(r) = \frac{N(r) - N((1-b)r)}{b N(r)}, \quad (4)$$

which tends to E as $b \rightarrow 0$ and equals $2 - b$ (not 2) under uniform density; with the default $b = 0.1$ the uniform reference is 1.9. Samples with fewer than a floor of members (30 by default) are flagged unreliable, because eq. (4) is a ratio of small counts.

$E(r)$ does not depend on the radius currently set, so it is computed once per centre and drawn on the radius slider itself: the analyst sees where the cliffs are before dragging onto one (fig. 2). This is an instance of embedding a visualization in the control that it informs, in the manner of scented widgets [67].

We do not claim the statistic. Counting within growing distances of a point is the idea behind Ripley’s

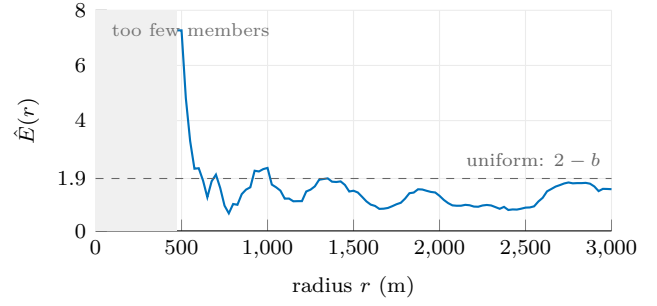


Figure 2: The elasticity profile $\hat{E}(r)$ (eq. (4), $b = 0.1$) around the demonstration centre in Yogyakarta, computed from the bundled OpenStreetMap extract by the library and plotted from its output. The shaded span has fewer than 30 members and is not drawn. The early cliff ($\hat{E} \approx 7.3$ at 475–500m) says a band of places at that distance is far denser than anything inside it, so a radius chosen there is fragile; a second, weaker rise follows at 925–1000m; and beyond about 1.4km the curve stays below the uniform reference, meaning that widening the lens adds ground faster than it adds places.

K function, for which $K(t) = \pi t^2$ under a Poisson process [49], and behind second-order neighbourhood analysis, which takes that view from each individual point [27]; under complete spatial randomness $N \propto r^2$ and so $E = 2$. The same log-log slope around a freely chosen centre is the mass-radius (fractal) dimension of radial analysis in urban morphology [68]. Three things distinguish the lens’s use of it: the centre is an arbitrary location rather than an event, so this is a point-to-event rather than an event-to-event count; the uniform reference depends on the lens shape (it is 2 for a disc or sector, but not for an annulus with fixed inner radius, nor 2 for a corridor whose width is the control); and it is shown as a property of the control rather than of the result. [TODO Report E relative to the selection’s own area elasticity, $E/(d \ln A/d \ln r) = \lambda_{\text{edge}}/\bar{\lambda}$, which is 1 under uniform density for every shape; and add a Monte-Carlo envelope under a null model [66] instead of the bare reference value.] [? The equivalences are derived here. Ripley and Yamu et al. were read in full; Getis & Franklin at abstract level only. Yamu et al. is mainly a planning-theory paper that uses radial analysis; a methods paper by Frankhauser would be the more canonical citation for mass-radius dimension (not yet identified and checked).]

4 Existing techniques as paths through the stages

A model of this kind is useful to the extent that it describes existing work without distortion, discriminates between techniques that differ, and suggests ones that

do not yet exist [3]. Table 1 codes techniques from the literatures of section 2 into the seven stages. Three observations follow from it.

Placement and anchor discriminate where the other stages do not. Excentric labelling, focus-region labelling, orbital labelling, the time lens and a category ring all select a disc; they differ in what they put around it and where. Without φ and γ as separate stages they would collapse into “a lens with labels (or charts) around it”.

Almost nothing places aggregates by bearing on a movable lens. Labelling techniques tie individual labels to bearing and resolve conflicts by dropping them; rings that aggregate order their marks by category or time, or space them evenly; necklace maps place one symbol per region by bearing, but around a whole map and without leaders. Among the techniques coded, the cells in which binning is by bearing *and* placement honours that bearing *and* the selection is a movable lens are occupied only by this work.

The tessellated techniques leave the join empty. GW maps, gridded glyphmaps, Honeycomb Plots and Hex-Tiles draw each summary *in place*; they have no placement problem because they have no boundary to place on. That is what makes the continuum possible: a field of lenses whose rings shrink towards their centres converges on the in-place case.

[TODO] This table is a single-coder, provisional coding for discussion among the authors. Before submission: fix the corpus by a recorded search protocol, double-code it (two coders, disagreements resolved and reported), and extend it to a corpus large enough to carry the claims (the lens design space we build on coded 45 papers [45]). A recent analysis of visualization design-space papers proposes a five-phase construction process—exploration, data collection, creation, evaluation, communication—that the final version should follow [15] (only its abstract has been read). Cells marked [?] rest on abstracts or are inferred rather than stated in the source; see the citation audit.]

5 New points in the space

Coding existing techniques shows what the model can *describe*. Its *generative* power [3] is the other half: whether it reaches configurations that were not built before, by changing arguments rather than by writing new techniques. Figure 3 shows eighteen configurations of one implementation on one dataset—1,449 OpenStreetMap places in central Yogyakarta, in four categories—each captioned with its path through the stages. We discuss the ones we believe are new, grouped by the stage that makes them possible. [“New” means: not found in the literature reviewed for section 2. Keep this wording until a systematic search (with a recorded protocol) backs a stronger one.]

Angle as bearing. Binning by bearing sector and placing each sector’s mark at its bearing (fig. 3c,f) turns a ring from a legend into a compass: a mark at eleven o’clock means the data it summarises lies to the north-west. The same move applied to categorical bins (fig. 3a→b) interpolates, with one parameter, from the familiar category ring to a bearing-faithful one; the transition is a lerp of preferred positions through the same solver. As section 8 discusses, the categorical version is honest only for segregated categories, and crossing bearing with category (fig. 3g) is the robust form.

The anchor as an axis. Because placement is solved in the curve’s parameter and holds length constant, the ring unrolls into a straight axis with no re-solve (fig. 3m), and a corridor straightens onto its own chainage in exactly the same way (fig. 3k→o). The unrolled ring is a bar chart with bearing on the horizontal axis—comparable lengths, room for labels—and it is a strip, so several lenses can share one baseline. Upright marks on a closed ring (fig. 3n) buy some of the same comparability without leaving the map.

Association as a residual. Leaders appear exactly where adjacency has been given away: under block placement, where every mark is far from its data (fig. 3l), and under unrolling, where the whole chart has left the map (fig. 3p). They are drawn to the members’ mean position in the lens’s own frame, so their length is the association lost.

Within-unit structure from the lens’s own geometry. The spread arc (fig. 3d) draws each category’s circular standard deviation, so a bar whose direction is not worth believing says so; rose marks (fig. 3e) replace the mark with its full angular distribution; and inclusions (fig. 3r) put the members back through the aggregate. On a corridor the same layer becomes lateral: bias and sidedness of member offsets.

5.1 The continuum from lens to glyphmap

Treating the number of lenses as a parameter gives a path from focus+context to small multiples to a gridded glyphmap (fig. 4). The field is not a second implementation: it calls the single-lens pipeline once per lattice centre and shares one baseline across all of them. Its lattice is itself an axis (hexagonal, square, triangular, or relaxed inside a boundary; fig. 3q), with a measurable consequence for how much ground the lenses reach (section 8).

Two findings from building it are worth stating. A field needs *one* baseline: a location quotient computed against each lens’s own surroundings makes every cell

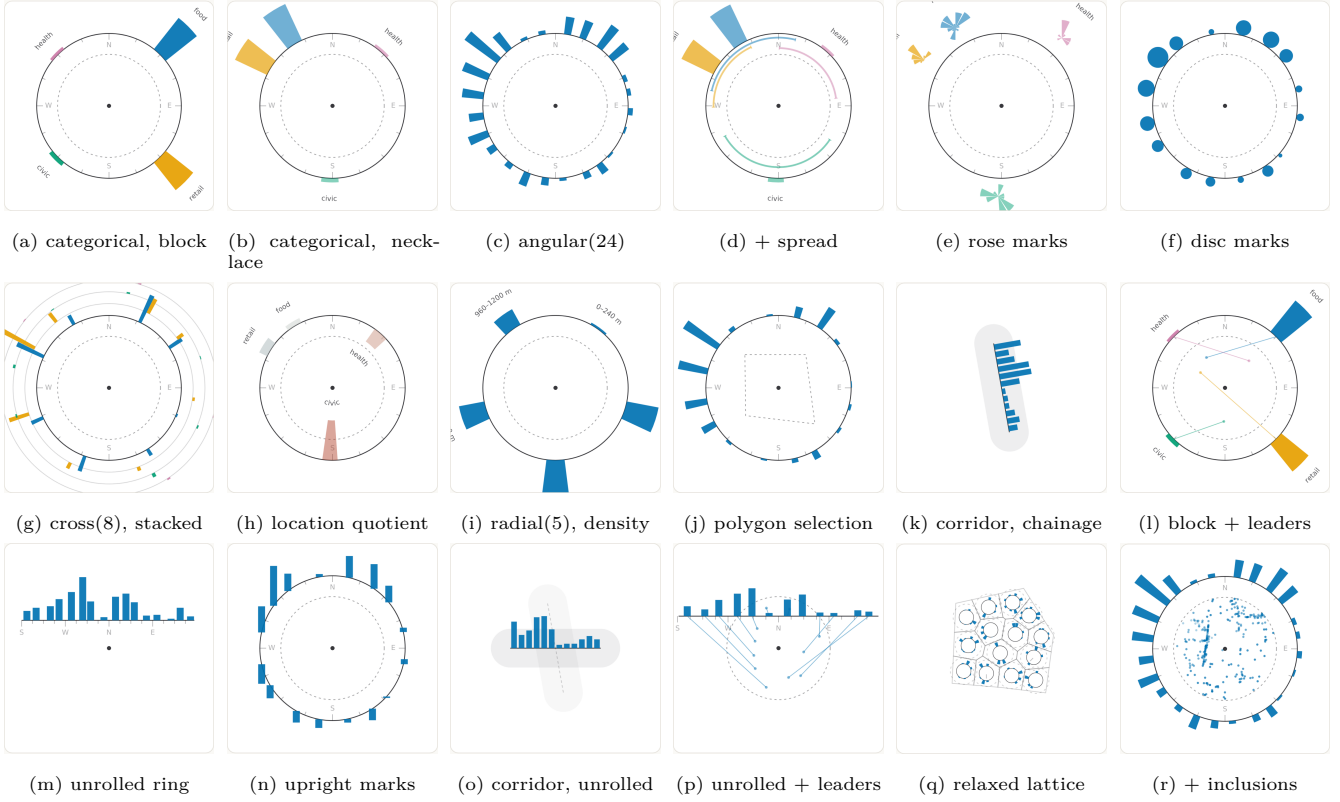


Figure 3: Eighteen configurations of one pipeline on one dataset (1,449 OpenStreetMap places, central Yogyakarta, four categories), rendered by the library on bare canvases with no map. (a) is the category ring common in lens demonstrations: angle carries category order. (b) is the same four bins placed at their members’ mean bearing. (c)–(f) bin by bearing sector and vary the mark and within-unit encoding. (g)–(k) vary binning, normalisation and selection. (l)–(p) vary the anchor and association. (q) is a field on a relaxed lattice inside a concave boundary; (r) draws the members back through their aggregates. Paths through the stages are given in the text and in the repository’s gallery.

average by construction, so the map says nothing. And the continuum has a usable range rather than an infinite one: as rings shrink the renderer sheds labels, then compass, and at the dense end the field reads as a surface (fig. 4, right).

The tessellated end is not new—it is the gridded glyphmap [40, 51, 52], and earlier tilemap work already proposed interactively resizing and panning the grid to expose the modifiable areal unit problem, and smoothing with a distance-decay kernel larger than the tiles [51]. What the model adds is that the single lens, the small multiple and the glyphmap are one object whose every other stage—binning by bearing, placement, association, within-unit structure—survives the change of count.

6 Implementation

glyphlens is an open-source JavaScript library (MIT licence) [TODO repository is private; make it public or deposit an archived snapshot (e.g. Zenodo DOI) before submission, and cite that]. The core is a set of pure

functions over plain objects with no dependency on a map, the DOM or a framework; it returns geometry, which a canvas renderer paints and a MapLibre adapter positions on a map. The seven stages of section 3 are the API: a lens is configured by one object per stage, and every figure in this paper is a single call with different arguments.

The core has no map in it. The gallery from which fig. 3 is drawn runs the whole pipeline onto bare canvases with no map library on the page. It doubles as a standing test that the core has not acquired a rendering dependency. All raster figures in this paper are regenerated from the library by a script in the repository (paper/scripts/figures.mjs), which serves the repository, renders the gallery and a figure page in headless Chromium, and saves each canvas.

Placement. The necklace solver of section 3.5 is independent of the rest of the library and exported on its own. It solves the fixed-order, interval-free case exactly by weighted pool-adjacent-violators, tries every cut of the

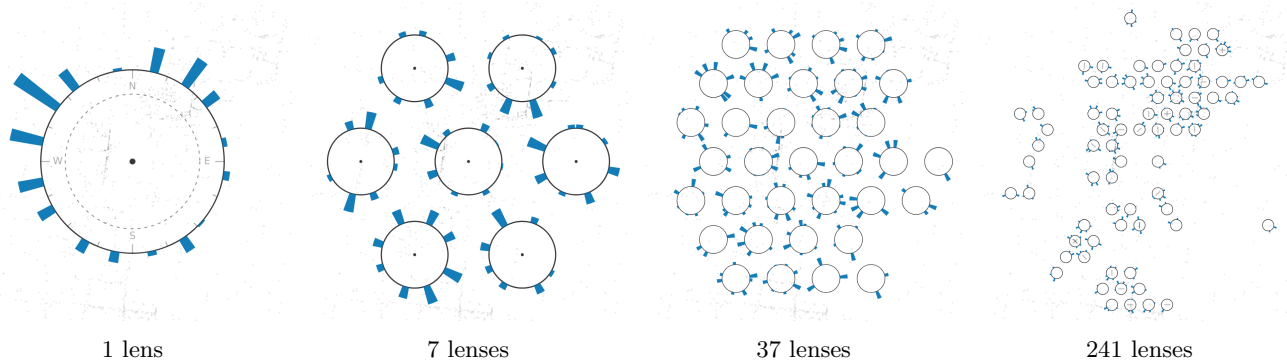


Figure 4: One parameter—the number of lattice centres—takes a single bearing-faithful lens to a gridded glyphmap. Every panel is the same binning (bearing sectors), the same solver and the same renderer; only the lattice changes. Lenses with fewer than three members are not drawn. Grey dots are the members.

cycle, and treats the nearly-full and interval-constrained cases approximately, as described there. Speckmann and Verbeek’s reference implementation is part of *CartoCrow*, a C++ library [? cite *CartoCrow* by its repository or project URL, with access date; confirm there is no associated paper to cite instead]. As far as we could find, there is no earlier JavaScript implementation [? “as far as we could find” is a search claim: record where we searched (npm, GitHub, Observable) and when].

Fields. A field calls the single-lens pipeline once per lattice centre and changes nothing else. It adds two things a single lens does not need: a lattice (hexagonal, square, triangular, or a Lloyd-relaxed centroidal Voronoi tessellation inside a boundary, built on a Delaunay triangulation) and a uniform grid index so that m lenses over n members do not cost $O(nm)$ distance tests.

Of the evaluation strategies used for toolkit research—demonstration, usage, technical benchmarks and heuristics [41]—this paper uses the first and third: the gallery and the coding of sections 4 and 5 are demonstrations, and table 2 is a benchmark. A usage study with developers is not part of this work.

Tests and performance. The repository has 196 unit tests (Node’s built-in runner, no runtime dependencies), covering among other things non-overlap, stability and overflow of the solver. Table 2 reports median wall-clock times for the pure pipeline, without rendering, on the bundled extract of 1,449 OpenStreetMap places.

The field of 889 lenses is no slower than the field of 241 because 804 of its lenses (and 165 of the 241) hold fewer than the minimum of three members and are skipped after the index query, so the cost is dominated by the lenses actually drawn (85 and 76 respectively). The solver is $O(K^2)$ in the number of marks, which is immaterial at the bin counts a lens uses and visible at 288.

7 Evaluation plan

[**TODO** This section describes a study that has *not* been run. No result in this draft comes from participants. Replace with the study once it is pre-registered and run; until then the core claim of section 3—that bearing-faithful placement helps—is a design conjecture and must be worded as one everywhere.]

The model’s descriptive and generative power is argued in sections 4 and 5; its central design claim is empirical and untested. Two of its axes make testable predictions, and they interact, so we plan one controlled, within-subjects study crossing them.

Factors. *Angle*: categorical (block placement, angle carries category order) versus bearing (necklace placement, angle carries bearing). *Anchor*: closed ring versus unrolled axis with leaders. A third, between-trials factor manipulates *displacement*—the gap between a mark and its true bearing, at four levels from 0 to 45 degrees [**TODO** levels to be fixed by pilot]—to locate the point at which a displaced mark misleads (the open policy question of section 3.8).

Tasks. Five tasks, each with a ground truth computable from the stimulus:

1. **Direction**: “In which direction from the centre is category X most concentrated?” Response: a bearing; error in degrees.
2. **Isotropy**: “Is X spread evenly around the centre?” Ground truth from the resultant length R .
3. **Comparison**: “Which of two highlighted directions has more X ?” Accuracy.
4. **Composition**: “Which category is most common inside the lens?” Accuracy. The categorical ring should do at least as well here; the task is included so that a win on direction is not bought silently.

5. **Association:** “Click the part of the map that the highlighted mark summarises.” Distance error.

Hypotheses. H1: bearing-faithful placement reduces direction error (task 1) relative to categorical placement. H2: categorical placement is no worse than bearing placement for composition (task 4). H3: the unrolled axis improves comparison accuracy (task 3) relative to the ring. H4: the ring beats the unrolled axis on association (task 5) without leaders, and leaders close the gap. H5: direction error grows with displacement, and leaders flatten that growth.

Stimuli, participants, analysis. Synthetic point patterns with controlled anisotropy (so ground truth is exact) plus a held-out set from real OpenStreetMap extracts, rendered by the library. Participants recruited online [TODO platform, *N* from a power analysis on pilot data, compensation, ethics approval]. Mixed-effects models with participant and stimulus as random effects [TODO specify before data collection; pre-register (e.g. OSF)].

Known risk. The closest empirical evidence cuts against direction-preserving placement. In a study of off-screen proxies on a rectangular viewport border, 18 participants’ localisation judgements were nearer the orthographic than the radial back-projection in 94.4% of cases, with lower error (78 versus 183 pixels), and 17 of 18 found orthographic more intuitive [36]. Two things limit the transfer: on a circular boundary the radial direction is the boundary’s own normal, and the authors themselves note that radial projection is the familiar one when there is a point of interest to project from, as a lens centre is. H1 and H5 are the direct test of whether bearing-faithful placement nonetheless helps on a lens. Relatedly, a study of orbital boundary labels around a circle ($n = 54$) found straight leaders faster than orbital-radial ones at similar accuracy [63]; our leaders are straight. [? Wallinger et al.’s numbers are from arXiv v1, not the PacificVis version of record.]

8 Discussion and limitations

A mean bearing is not always a direction. Positioning a category’s mark at the circular mean of its members is only honest when they are concentrated. On the bundled demonstration data (315 places within 1.2km of the centre) the two large categories point the same way as the data as a whole: retail at 302° ($R = 0.51$, $n = 135$) and food at 329° ($R = 0.28$, $n = 153$), against 313° ($R = 0.34$) for all places. Health points north-east (42° , $n = 11$) and civic south (182° , $n = 16$), but civic’s resultant length is 0.12, so its mean bearing is close to arbitrary. A bearing-faithful ring

therefore separates the large categories by concentration rather than by direction, and gives up the composition reading the categorical ring gave for free. (An earlier extract recorded in the development log showed all four means within about twenty degrees; which data it was is not recorded.) The morph from category order to bearing is strongest for spatially segregated categories. Two responses are built: marks are faded by their resultant length R , and the *cross* binning (bearing $\times$ category) gives each category one mark per sector, so that a diffuse category shows as a spread of marks rather than one misplaced one. We regard the second as the honest multivariate view and the first as a limitation to be stated, not a fix.

How multivariate is this? Most configurations in this paper summarise one categorical variable by count. Several variables are supported through cross binning, stacked necklaces (one per variable, exact angular fidelity at the cost of radial space), and areal measures with declared extensive or intensive semantics; but we have not shown that these scale beyond a handful of variables, and the glyph literature suggests they will not without careful design [10, 25]. [TODO Either demonstrate a genuinely multivariate census case (e.g. several deprivation domains) or narrow the claim in the title and abstract.]

Relation to geographically weighted statistics. A field of disc-shaped lenses on a lattice computes, at each centre, local summaries with a uniform (hard cut-off) kernel—geographically weighted summary statistics [11] with the simplest kernel. Dykes and Brunson call the GW mean “simply a moving spatial window mean smoother” and note that the statistics could be centred on the points of a regular grid [19], and a field drawn as glyphs is what tilemaps of GW outputs were proposed to be [51]. We do not claim the statistics. What the lens adds is (i) keeping the within-lens decomposition by bearing and distance, drawn at its true bearing, where a GW summary is one scalar per location, and (ii) one model in which the single lens and the field are the same object at different settings. [TODO Implement bi-square and Gaussian kernels so that the field reproduces GW summary statistics exactly, and show it; this turns the obvious reviewer objection into a validity check.] [? Brunson et al. 2002 was read at abstract level; whether it discusses a uniform kernel explicitly is unverified. Dykes & Brunson and Slingsby 2018 were read in full.]

The continuum has a usable range. As the lattice densifies each ring shrinks, and below roughly ten pixels of ring radius a glyph stops carrying multivariate information and the field reads as a density surface with texture [? an observation from building the demonstrations,

not a measured threshold]. This is the trade between spatial and multivariate resolution that any glyph map makes; the renderer sheds labels and compass as rings shrink, but the threshold itself is a candidate for the study.

Lenses do not tessellate. Centres on a lattice tessellate the plane; the discs they select do not. At touching packing (disc radius half the nearest-neighbour spacing) a hexagonal lattice covers $\pi/(2\sqrt{3}) \approx 90.7\%$ of the ground, a square lattice $\pi/4 \approx 78.5\%$ and a triangular (honeycomb) lattice $\pi/(3\sqrt{3}) \approx 60.5\%$. Any larger disc makes neighbours overlap and count some members twice, while on a hexagonal lattice the gaps do not close until the disc radius reaches the cell’s circumradius, $s/\sqrt{3}$: between $s/2$ and $s/\sqrt{3}$ a field both misses ground and double-counts it, so no disc radius is both gap-free and overlap-free. Drawing only the cell would be the comfortable lie, so the field reports coverage and can draw the disc, the cell, or both.

Other limitations. Bearing from a polygon’s centroid is ambiguous for elongated or concave units, where the population-weighted centroid, the geometric centroid and the unit’s full angular extent disagree. The baseline for the location quotient (complement, region, nation) changes the reading and should be visible, not a configuration default. Placement is exact only in the common case (section 3.5). Density denominators come from metric radii and, for polygons, a local equirect-angular frame, never from Web Mercator screen space, which inflates area by $1/\cos^2 \phi$ at latitude ϕ .

Name. The library’s working name collides with an unrelated occlusion-management technique for 3D glyph fields [60]; it will be renamed before publication.

9 Conclusion

A lens that summarises the data under it has to answer two questions that a magnifying lens does not: what to summarise, and where to put the summary. We split Tominski et al.’s lens function and join into three stages each, and showed that in the resulting seven-stage space a category ring, a bearing-faithful necklace, a route profile and a gridded glyphmap are points reached by changing arguments rather than by writing new techniques. The part of the space we find most interesting is where the lens boundary itself carries data: a ring whose angle means bearing, whose marks are tied back to the ground by leaders when placement or unrolling gives adjacency away, and whose radius control shows how fragile the reading is. Whether that helps anyone is the question the planned study exists to answer.

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Table 1: Provisional coding of existing techniques into the seven stages. “—” means the stage is absent or trivial; [?] marks a code that is inferred or rests on an abstract only.

Technique	σ selection	β binning	ν normal.	φ placement	γ anchor	μ marks	ρ association
Magic lens [6]	screen region	—	—	in place	interior	restyled objects	co-location
Excentric labelling [21]	disc at cursor	one per object; count when too many	—	ordered stacks (radial variant: by bearing)	two columns beside the disc	labels	lines
Focus-region labelling, radial model [22]	disc	one per site (or per cluster)	—	port at site’s bearing; conflicts <i>dropped</i>	the circle	labels	radial leaders, always
Orbital boundary labelling [9]	disc	one per feature	—	minimise total leader length	annulus around the circle	arc labels	straight or orbital-radial leaders
Probes [13]	disc or units	per chart	per chart	placed by user	separate pane	charts	line or shared colour
Time lens [57]	disc (query circle)	cyclic time bins	count, duration	time order	ring around the query [?]	filled ring segments	adjacency
VisQuill lens [62]	disc	categorical	count [?]	evenly spaced, category order [?]	ring	radial bars	adjacency
Route detail lenses [37]	route neighbourhoods	one per point of interest	—	border slots, route order, direction-coherence cost	map frame	magnified views	first/last leaders, arrows
ClusterLens [70]	lens	clusters or grid at finer resolution	count	in place	interior	heatmap, circles, grid	co-location
Necklace map [53]	whole map (regions)	one per region	value	max common scale in intervals, then centring	star-shaped curve around the map	scaled discs	proximity, colour; no leaders
Ring map [55]	whole map (units)	one spoke per unit	attribute classes	evenly spaced spokes	concentric rings around the map	coloured spoke segments	leader for every spoke
CompaRing [56]	user-chosen objects (< 10)	one per object	—	ring slots (not by direction)	ring	object copies	arc toward the original, width $\propto$ distance
Ambient Grids [35, 36]	off-screen space	border grid cells	count or density [?]	orthographic [?]	viewport border	heatmap cells	position
VectorLens [18]	disc + direction range	(selection only)	—	—	ring as control	selected angular ranges	—
GW summary statistics [11, 19]	kernel at each location; directed kernel at clock points	— (or by direction)	weighted mean, SD, quantiles	in place	—	colour	co-location
Gridded glyphmap [40, 52]	screen-space grid cell	variables	per variable	cell centre	—	glyph	position
Honeycomb plot [61]	hexagon	one, plus within-tile gradient	density	in place	—	height, diamond cut	position
HexTiles [38]	hexagon	variables	+ variance-based confidence	in place	—	colour, icons, ring	position
This work: bearing ring	disc, annulus, sector, corridor, polygon	bearing sector (or cross, unit, chainage)	count, density, share, LQ, z, delta	necklace (minimal displacement)	ring $\rightarrow$ axis ($u \in [0, 1]$)	bar, disc, rose	adjacency; leader where the gap misleads
This work: field	disc per lattice centre	as above	shared baseline	per lens	per lens	as above	as above

Table 2: Median time of the pure pipeline (no rendering), Node 22 on one core of a 2.8 GHz Intel Xeon, 1,449 places; produced by `paper/scripts/bench.mjs`. Field sizes are the actual lattice counts.

Configuration	ms
One lens, 8 bearing sectors, necklace placement	0.9
One lens, 24 bearing sectors	1.6
One lens, 72 bearing sectors	5.2
One lens, 8 sectors $\times$ 4 categories, stacked	0.8
Field of 37 lenses, 8 sectors each	5.4
Field of 241 lenses	16.3
Field of 889 lenses	15.5
Solver alone, 24 marks, 80% fill	0.6
Solver alone, 72 marks	1.1
Solver alone, 288 marks	15.4

Appendix: citation audit (remove before submission)

Every entry in the bibliography was checked on 2026-09-23 and 2026-09-25 against its registry record—Crossref, or DataCite for the Eurographics (10.2312), Dagstuhl (10.4230) and arXiv (10.48550) DOIs—or, where there is no DOI, against the page at its URL. The table records, for each source, the evidence behind what this draft says about it: **F** read in full (“pre” marks a preprint or author copy rather than the version of record), **A** abstract only, **M** bibliographic record only, **W** web pages. Verbatim quotes with page references are in `paper/citation-audit.md`. `paper/scripts/check-bib.mjs --online` repeats the registry check; on 2026-09-25 all 67 DOIs resolved and every registered title matched.

Source	Ev.	Used for	Caveat
<i>Lens models and design-space method</i>			
Bier et al. 1993 [6]	F	origin of magic lenses	—
Bier et al. 1994 [7]	F	taxonomy of see-through tools	a taxonomy of see-through tools in general, not of lenses; several same-title DOIs exist
Fox 1998 [24]	F	composing lenses	a programming-model contribution (delegation), not a visual one
Tominski et al. 2014 [58]	F	$\sigma/\propto$ model; data types and tasks; future directions (combination, toolkit)	the symbol λ is not in the 2014 STAR; neither survey lists adaptive lenses or evaluation as future directions
Tominski et al. 2017 [59]	F pre	λ notation; effect extent (interior, side effects, separate view)	online 2016, issue 2017
Mota et al. 2025 [45]	F	seven dimensions; effect scope and encoding values; 45-paper corpus	arXiv v2 only; no published version found
Dai et al. 2026 [15]	A	five-phase construction process	full text not accessed; nothing about how it evaluates design spaces is used
Beaudouin-Lafon 2004 [3]	F	descriptive, evaluative, generative power	—
Ledo et al. 2018 [41]	F	toolkit evaluation strategies	—
<i>Lenses and probes on maps</i>			
Tong et al. 2017 [60]	A	name collision; occlusion lens for 3D glyphs	—
Hurter et al. 2011 [34]	F	semantic lens that pushes elements to its border	author order is Hurter, Ersoy, Telea (several surveys have it wrong)
Pindat et al. 2012 [48]	F	content-adaptive lens shape	—
Zhang et al. 2020 [70]	F	re-aggregation inside a lens	no charts or ring
Karnick et al. 2010 [37]	F pre	detail lenses on the map border in route order; leaders to first and last only; direction-coherence cost	preprint
Ellis et al. 2005 [20]	F	sampling lens; aggregates as future work	scatterplots and parallel coordinates, not maps
Dumas et al. 2015 [18]	F	ring encoding direction for selection	online 2014, issue 2015; no map example
Butkiewicz et al. 2008 [13]	F	probes: region + user-placed pane of charts, linked by line or colour	—
Butkiewicz et al. 2010 [14]	F pre	MAUP alerts between compared regions; scale component explicitly excluded	read the anonymised EuroVis submission, not the published version
Krüger et al. 2013 [39]	A	set-operation lenses; aggregates near the lens or in colour-matched panels	full text not accessed (open copy withdrawn); verbatim related-work excerpts only
Scheepens et al. 2016 [50]	A	selection widget by area and direction; annotation windows	whether it is called a lens, and where windows sit, unverified
Ma et al. 2020 [42]	A	movable lenses over geo-text	what is shown and where, unverified
Tominski et al. 2012 [57]	F pre	time lens: ring of cyclic time bins around a query circle	ring placement around the circle strongly implied, not stated
VisQuill 2026 [62]	W	shipped aggregating ring lens on maps	angle meaning not documented; Lab blueprint spaces one bar per category
<i>Labelling around a focus region</i>			

Source	Ev.	Used for	Caveat
Fekete and Plaisant 1999 [21]	F pre	left/right label stacks; radial variant orders by bearing; summary glyph as future work	read the HCIL tech report; its study numbers differ from how others report them—do not quote them
Bertini et al. 2009 [5]	A	density extensions; “summary statistics”	what the statistics are is unverified
Fink et al. 2012 [22]	F pre	labels on the circle; radial ports; conflicts dropped; clustering variant; focus placement	—
Haunert and Hermes 2014 [32]	A	leaders to the centre; one position per label; real-time JavaScript	—
Heinsohn et al. 2014 [33]	A	moving focus regions	mechanism for coherence unverified; biological networks
Niedermann and Haunert 2019 [46]	A	fish-eye displacement; optimisation when the focus stops	program type unverified
Bekos et al. 2019 [4]	F pre	external-labelling taxonomy	arXiv v2
Bonerath et al. 2024 [9]	F	orbital arc labels; minimum leader length	—
Wallinger et al. 2026 [63]	F pre	$n = 54$; straight leaders faster, similar accuracy	numbers from arXiv v1, not the PacificVis version
Gedicke et al. 2021 [26]	F pre	labels along the map’s lower edge	vol. 27 (2021) despite a 2020 DOI
<i>Glyph maps and within-cell structure</i>			
Ward 2002 [64]	F pre	data-driven vs structure-driven placement	author manuscript
Borgo et al. 2013 [10]	F	glyph design guidance	DataCite lists 2012 in error; pages from the DOI suffix
Fuchs et al. 2017 [25]	F pre	64 studies; layout and map backgrounds understudied	—
Wickham et al. 2012 [65]	F pre, A	glyphs at the data’s own grid locations	quote the version of record’s abstract where possible
McNabb and Laramee 2019 [44]	A, F pre	hierarchical-centroid placement with LOD	—
Slingsby 2018 [51]	F	tilemaps; GW outputs <i>proposed</i> ; grid resizing and larger kernels discussed	workshop paper, no DOI; GW use not demonstrated
Slingsby et al. 2023 [52]	F pre	screen-space grid re-aggregated on zoom; within-cell heterogeneity as future work	—
Laksono et al. 2024 [40]	F	gridded glyphmaps for GMCD	no user study
Trautner et al. 2022 [61]	F pre	diamond cut; $n = 42$ study; the C3 sentence	the quoted sentence concerns colour-only hexbins; amber inclusions untested
Kawakami et al. 2024 [38]	F	variance-based per-tile confidence	arXiv only; no formula; study ($n = 7$) without the variance encoding, no significant effects
Boeing 2019 [8]	F pre	polar histograms of street bearings	bearing of edges, not from a centre
Gleicher et al. 2011 [28]	A	juxtaposition, superposition, explicit encoding	—
<i>Local statistics, scale and point patterns</i>			
Brunsdon et al. 1996 [12]	A	origin of GW methods	the “sudden cut-off” reading is unverified and not used
Brunsdon et al. 2002 [11]	A	GW summary statistics	whether it names a uniform kernel is unverified
Dykes and Brunsdon 2007 [19]	F pre	“moving spatial window mean smoother”; scalograms; regular grid; directed weighting	Gaussian kernel; h precomputed, not a slider; check the sign in the directed-kernel formula before reproducing it
Goodwin et al. 2016 [29]	F pre	local correlation across scale	—
Anselin 1995 [1]	A	local indicators	—
Openshaw 1984 [47]	F (scan)	MAUP: scale and aggregation problems	reviews and defines the problem, does not introduce it; year not printed in the scan

Source	Ev.	Used for	Caveat
Fotheringham and Wong 1991 [23]	A	MAUP unpredictable in multivariate analysis	—
Ripley 1977 [49]	F	$K(t) = \pi t^2$ under Poisson; envelopes	the L -function is not in the paper body
Getis and Franklin 1987 [27]	A	point-centred second-order analysis	the L_i formula is unverified and not used
Wiegand and Moloney 2004 [66]	A	circle vs ring statistics; null models	“envelopes” not verified in this paper
Yamu et al. 2016 [68]	F	radial (mass–radius) analysis	a planning-theory paper; a methods citation would be better
Willett et al. 2007 [67]	F	scented widgets	credits histogram sliders to earlier work
Mardia and Jupp 1999 [43]	M	circular mean, resultant length, circular SD	book not accessed; definitions are standard but the equation numbers are unverified
<i>Off-screen indicators, necklace maps, ring maps</i>			
Speckmann and Verbeek 2010 [53]	F pre	regions projected onto intervals; symbols without overlap; multiple necklaces; max scale then centring; “do not need leaders”	—
Speckmann and Verbeek 2015 [54]	F pre	fixed order $O(n \log n)$; any order NP-hard (wedge intervals); bearing order not always optimal for max scale	accepted manuscript; objective differs from ours
Stewart et al. 2011 [55]	F	evenly spaced spokes, one per unit; a leader for every spoke	ordering rule of spokes not verified
Tominski 2016 [56]	F pre	ring slots with arcs toward originals; direction-ordered slots only suggested	—
Jäckle et al. 2017 [36]	F pre	$n = 18$; 94.4% orthographic; radial familiar with a point of interest	arXiv v1 only; rectangular border
Baudisch and Rosenholtz 2003 [2]	M	off-screen proxy (via Jäckle et al.’s classification)	own text not read
Zellweger et al. 2003 [69]	M	off-screen proxy (via Jäckle et al.)	own text not read
Gustafson and Irani 2007 [31]	M	border region (via Jäckle et al.)	own text not read
Gustafson et al. 2008 [30]	M	off-screen proxy (via Jäckle et al.)	own text not read
Jäckle et al. 2015 [35]	M	aggregated border grid (via Jäckle et al.)	own text not read
Danyluk et al. 2026 [16]	A (partial)	AR ring maps for pedestrian navigation	how landmarks are placed is unverified
Draper et al. 2009 [17]	M	survey of radial methods (its title)	—